L'Hôpital's Rule

Questions 1–18

Question 1. Use L'Hôpital's rule to find:

a) $\lim_{x\to 2}\left(\dfrac{x^2-4}{x-2}\right)$

b) $\lim_{x\to 1}\left(\dfrac{\ln x}{x-1}\right)$

c) $\lim_{x\to 0}\left(\dfrac{e^x-1}{x}\right)$

d) $\lim_{x\to 0}\left(\dfrac{e^{2x}-1}{x}\right)$

e) $\lim_{x\to 0}\left(\dfrac{e^x-1}{\ln(1+x)}\right)$

f) $\lim_{x\to 0}\left(\dfrac{\ln(1+3x)}{x}\right)$

Question 2. Use L'Hôpital's rule to find:

a) $\lim_{x\to 0}\left(\dfrac{\sin x}{x}\right)$

b) $\lim_{x\to 0}\left(\dfrac{1-\cos x}{x}\right)$

c) $\lim_{x\to 0}\left(\dfrac{x}{\sin x}\right)$

d) $\lim_{x\to 0}\left(\dfrac{x^2}{\sin x}\right)$

e) $\lim_{x\to 0}\left(\dfrac{-x+\tan x}{x}\right)$

f) $\lim_{x\to 0}\left(\dfrac{\sin 3x}{x}\right)$

Question 3. Use L'Hôpital's rule to find:

a) $\lim_{x\to 0}\left(\frac{1-\cos x}{x^2}\right)$

b) $\lim_{x\to 0}\left(\frac{x-\sin x}{x^3}\right)$

c) $\lim_{x\to \infty}\left(\frac{\ln x}{x}\right)$

d) $\lim_{x\to \infty}\left(\frac{x^2}{e^x}\right)$

e) $\lim_{x\to 0}\left(\frac{e^{3x}-1-x}{x^2}\right)$

f) $\lim_{x\to 0}\left(\frac{\ln(1+x)-x}{x^2}\right)$

g) $\lim_{x\to \infty}\left(\frac{\ln x}{\ln(x+1)}\right)$

Question 4. Use L'Hôpital's rule to find:

a) $\lim_{x\to 0}\left(\frac{\sinh x}{x}\right)$

b) $\lim_{x\to 0}\left(\frac{\tanh x}{2x}\right)$

c) $\lim_{x\to 0}\left(\frac{\sinh x-x}{4x^3}\right)$

Question 5. Use L'Hôpital's rule to find:

a) $\lim_{x \to 0} \left(\frac{\arcsin x}{x} \right)$

b) $\lim_{x \to 0} \left(\frac{\arctan x}{x} \right)$

Question 6. A student tried to find the value of $\frac{\cos x + 1}{e^x + 2}$ as $x \to 0$. They wrote:

$$\lim_{x \to 0} \left(\frac{\cos x + 1}{e^x + 2} \right) = \lim_{x \to 0} \left(\frac{-\sin x}{e^x} \right) = 0.$$

a) Explain the student's mistake.

b) Find the correct value of the limit.

Question 7. Use L'Hôpital's rule to evaluate $\lim_{x\to\infty}(xe^{-2x})$.

Question 8.

a) Rewrite $x\ln(1+\frac{1}{x})$ as a quotient for the use of L'Hôpital's rule.

b) Hence, use L'Hôpital's rule to find $\lim_{x\to\infty}\left(5x\ln(1+\frac{1}{x})\right)$.

Question 9. Use L'Hôpital's rule to find:

a) $\lim_{x\to 0}\left(\dfrac{xe^x}{\sin x}\right)$

b) $\lim_{x\to 0}\left(\dfrac{e^x \sin 7x}{\ln(4x+1)}\right)$

c) $\lim_{x\to 0}\left(\dfrac{\sin x - x\cos x}{x^3}\right)$

d) $\lim_{x\to 0}\left(\dfrac{1-\cos 5x + 3x\sin 3x}{2x^2}\right)$

Question 10. Use L'Hôpital's rule to find:

a) $\lim_{x\to \pi}\left(\dfrac{\sin x}{x-\pi}\right)$

b) $\lim_{x\to \pi}\left(\dfrac{x\cos x + \pi}{(x-\pi)^2}\right)$

c) $\lim_{x\to \pi}\left(\dfrac{\cos(\frac{x}{2})}{x\sin 2x}\right)$

Question 11. Use L'Hôpital's rule to find:

a) $\lim_{a\to\infty}\left(\frac{a}{e^a}\right)$

b) Hence find $\int_0^\infty xe^{-x}\,dx$.

Question 12. Use L'Hôpital's rule to find:

a) $\lim_{a\to\infty}\left(\frac{\ln a}{a}\right)$

b) Hence find $\int_1^\infty \frac{\ln x}{x^2}\,dx$.

Question 13.

a) Show that $\dfrac{d}{dx}[\ln(\cos x)] = -\tan x$.

b) Hence use L'Hôpital's rule to evaluate $\lim\limits_{x\to 0}\left(\dfrac{\ln(\cos x)}{x^2}\right)$.

Question 14.

a) Show that $\dfrac{d}{dx}[\ln(\cosh x)] = \tanh x$.

b) Hence use L'Hôpital's rule to evaluate $\lim\limits_{x\to 0}\left(\dfrac{\ln(\cosh x) - 7x^2}{x^2}\right)$.

Question 15.

a) Use L'Hôpital's rule to show that $\lim_{x\to 0^+}(x\ln x) = 0$.

b) Explain why we must consider $x \to 0^+$ and not $x \to 0^-$.

Question 16. The displacement s of a particle is given by $s(t) = \dfrac{e^t - 1}{t}$. Use L'Hôpital's rule to find the velocity of the particle at $t = 0$.

Question 17. Given that, for $x \geq 0$, $f(x) = x^{-3}(\ln(1+x) - x + \frac{1}{2}x^2)$:

a) Use a Maclaurin expansion to show that $\ln(1+x) - x + \frac{1}{2}x^2 \approx \frac{1}{3}x^3$.

b) Hence deduce that $\lim_{x\to 0}(x^{-3}[\ln(1+x) - x + \frac{1}{2}x^2]) = \frac{1}{3}$.

c) Use L'Hôpital's rule to confirm the result in part (b).

d) Explain why L'Hôpital's rule alone is not sufficient to determine the behaviour of $f(x)$ near $x = 0$.

Question 18. By considering where $y = (1 + \frac{1}{x})^x$, use L'Hôpital's rule to find $\lim_{x\to\infty}(1 + \frac{1}{x})^x$.